

Software, Hardware, Everywhere

Programming begins in high school with Visual Studio.NET and MS Access. Algebra, that scary phenomenon, is tamed with Derive, while Geometer's Sketchpad opens up the monitor as a visualisation tool to help grasp the concepts of lines, planes and much more. Electronic book-keeping is taught using MYOB, and Environmental Science projects use Front Page Editor. Measuring data for experiments in physics, chemistry and biology are done with handheld devices—the results are uploaded onto a network folder, and analysis can be done from the comfort of one's dormitory using Vernier LabPro!

The use of software is not limited to the classroom. Publisher 2000 enables the newspapers brought out by the students in the journalism class, while the VHS editing room is used to compile an audio-visual news magazine, with the international section being sourced from CNN, BBC and local happenings shot around the campus. The yearbook, again produced in-house, is created using the Adobe DTP Suite.

Sibelius Orchestration and Finale Orchestration are tools used to teach music skills across all three sections—the kindergarten, middle school and high school. Not only do the software make provision for the students to write music, they offer exercises in creating "missing parts." For those looking to master improvisational techniques, Band-in-a Box allows one to "solo" behind a band. And after hours of practice, when a young aspirant wants to apply to music college, Sonar Recording Software helps cut that master.

305 workstations, 15 servers, a 1 terabyte Storage Area Network (SAN) set to scale up to 2 TB in the next two years, a robotic tape library, 2 Mbps Internet connectivity, and 14 km of fibre-optic cable criss-crossing a location challenged by lightning, rain, fog and monkeys—along with the associated hubs and switches—comprise the infrastructure that enables all this and more.

The Man Behind The Machines

Overseeing mission critical e-mail—often the only link for the community, gathered from 25 Indian states and 35 countries, with the outside world—is a job in itself. So is the smooth functioning of the BlackBaud ERP suite (See box *Blackbaud*) on which the entire school administration runs. There's also the backup of the enormous amount

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of data generated on the SAN to be managed. And there's one man behind all this—fifty-one-year old Steve Ediger, Director- ICT (Information and Communication Technology) at Woodstock.

Ediger, married to a Woodstock alumnus, came to Mussoorie nine years ago after selling his business in Los Angeles. A veteran in the field of IT right from 1978—when there were 50,000 micro-computers in the US—he started out as a distributor for systems and peripherals with a company that later became Ingram. Ediger remembers Bill Gates walking in and out of offices trying to peddle the compilers he had written prior to DOS!

When Ediger came to Woodstock in 1997, the school had 100 computers running multiple versions of DOS, Windows, and WordPerfect on a LAN. There were two Intel servers and a 64 Kbps Internet link shared between 300 users.

The Upgrade

The existing discrete storage method, where the two application servers also doubled up as storage, had its disadvantages. For example, teachers had to store both lesson plans and personal data on a single network drive. When a teacher left, someone would have to sift through all that teacher's data, pulling out what was needed by the teacher who came in. Also, from the system administrator's point of view, e-mail maintenance meant shutting down a common server, which would naturally hamper other applications running on it.

The use of IT in the classrooms was increasing: e-yearbooks began to take up entire drives. When the biology class completed study of a stream ecology, the data and pictures would take up another drive. And the Internet was being used much more as a research tool. All this meant information at Woodstock was exploding. And so, in January of 2004, plans to upgrade the infrastructure were made. The approach had to factor in e-mail, ERP, MIS and classroom requirements. At the end of the upgrade, Woodstock was truly wired. While the final cost came to more than Rs 3 crore, the initial costs included Rs 50 lakh on the servers, SAN and tape; Rs 10 lakh for the additional hardware; Rs 12 lakh for new software; and Rs 10 lakh for the 14 km of fibre-optic cable.

Ediger's comments on this venture are worth noting: "We believe we have investment protection because we can get even faster performance, continuous availability and data integrity, and increased scalability—from five to 240 drives in a modular, pay-as-we-grow increments. This allows me to plan not just for next year, but for at least the next five years, which is a true luxury when it comes to information technology."

"A Twenty-Fold Improvement"

For the residents of the school, the upgrade meant that each classroom got a computer, used primarily for presentation. The science rooms together got 100. There were 25 for each of the three labs, while the Internet library and the study room together got 18. The dormitories were not neglected: while the high school dorms managed a 1:10 ratio, middle-school dorms got a lower 1:15. Elementary school dorms, where the little residents have many other concerns, make do with a 1:50 ratio.



There's no dearth of computer lovers here